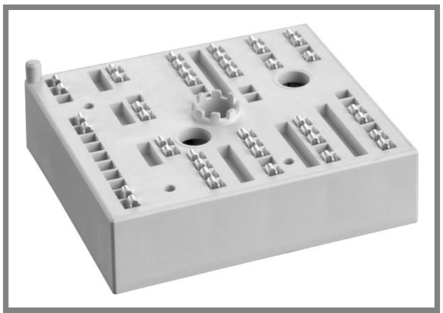


MiniSKiiP® 2

3-phase bridge rectifier + brake chopper + 3-phase bridge inverter

SKiiP 23NAB126V10

Preliminary Data

Features

- Fast Trench IGBTs
- Robust and soft freewheeling diodes in CAL technology
- Highly reliable spring contacts for electrical connections
- UL recognised file no. E63532

Typical Applications*

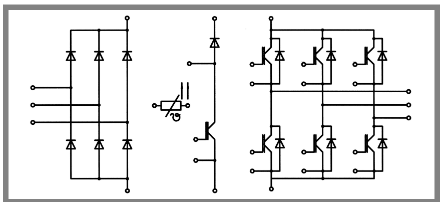
- Inverter up to 14 kVA
- Typical motor power 7,5 kW

Remarks

- V_{CEsat} , V_F= chip level value

Absolute Maximum Ratings		T_s = 25 °C, unless otherwise specified	
Symbol	**Conditions**	**Values**	**Units**
IGBT - Inverter, Chopper			
V_{CES}		1200	V
I_C	T_s = 25 (70) °C	41 (31)	A
I_{CRM}		50	A
V_{GES}		± 20	V
T_j		- 40 ... + 150	°C
Diode - Inverter, Chopper			
I_F	T_s = 25 (70) °C	30 (22)	A
I_{FRM}		50	A
T_j		- 40 ... + 150	°C
Diode - Rectifier			
V_{RRM}		1600	V
I_F	T_s = 70 °C	61	A
I_{FSM}	t_p = 10 ms, sin 180 °, T_j = 25 °C	700	A
i²t	t_p = 10 ms, sin 180 °, T_j = 25 °C	2400	A²s
T_j		- 40 ... + 150	°C
Module			
I_{tRMS}	per power terminal (20 A / spring)	40	A
T_{stg}		- 40 ... + 125	°C
V_{isol}	AC, 1 min.	2500	V

Characteristics			T_s = 25 °C, unless otherwise specified		
Symbol	**Conditions**	**min.**	**typ.**	**max.**	**Units**
IGBT - Inverter, Chopper					
V_{CEsat}	I_{Cnom} = 25 A, T_j = 25 (125) °C		1,7 (2)	2,1 (2,4)	V
$V_{GE(th)}$	V_{GE} = V_{CE}, I_C = 1 mA	5	5,8	6,5	V
$V_{CE(TO)}$	T_j = 25 (125) °C		1 (0,9)	1,2 (1,1)	V
r_T	T_j = 25 (125) °C		28 (44)	36 (52)	mΩ
C_{ies}	V_{CE} = 25 V, V_{GE} = 0 V, f = 1 MHz		1,8		nF
C_{oes}	V_{CE} = 25 V, V_{GE} = 0 V, f = 1 MHz		0,3		nF
C_{res}	V_{CE} = 25 V, V_{GE} = 0 V, f = 1 MHz		0,2		nF
$R_{th(j-s)}$	per IGBT		0,9		K/W
$t_{d(on)}$	under following conditions		85		ns
t_r	V_{CC} = 600 V, V_{GE} = ± 15 V		30		ns
$t_{d(off)}$	I_{Cnom} = 25 A, T_j = 125°C		465		ns
t_f	R_{Gon} = R_{Goff} = 30 Ω		100		ns
E_{on}	inductive load		3,5		mJ
E_{off}			3		mJ
Diode - Inverter, Chopper					
V_F = V_{EC}	I_{Fnom} = 25 A, T_j = 25 (125) °C		1,8 (1,8)	2,1 (2,2)	V
$V_{(TO)}$	T_j = 25 (125) °C		1 (0,8)	1,1 (0,9)	V
r_T	T_j = 25 (125) °C		32 (40)	40 (52)	mΩ
$R_{th(j-s)}$	per diode		1,7		K/W
I_{RRM}	under following conditions		33		A
Q_{rr}	I_{Fnom} = 25 A, V_R = 600 V		5,7		µC
E_{rr}	V_{GE} = 0 V, T_j = 125 °C		2,5		mJ
	di_F/dt = 1140 A/µs				
Diode - Rectifier					
V_F	I_{Fnom} = 35 A, T_j = 25 °C		1,1		V
$V_{(TO)}$	T_j = 150 °C		0,8		V
r_T	T_j = 150 °C		11		mΩ
$R_{th(j-s)}$	per diode		0,9		K/W
Temperature Sensor					
R_{ts}	3 %, T_r = 25 (100) °C		1000(1670)		Ω
Mechanical Data					
w			65		g
M_s	Mounting torque	2		2,5	Nm



NAB

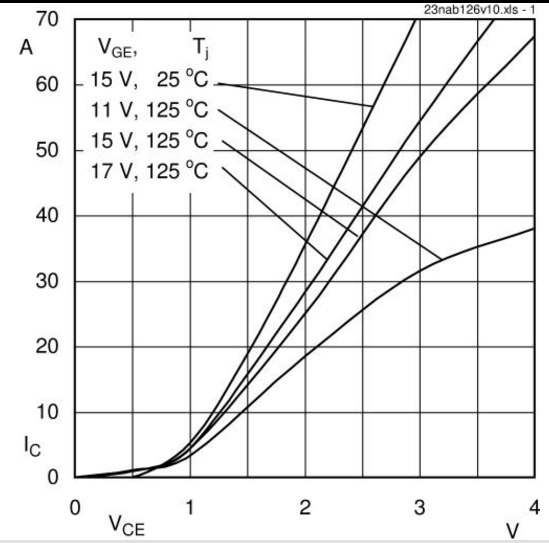


Fig. 1 Typ. output characteristic

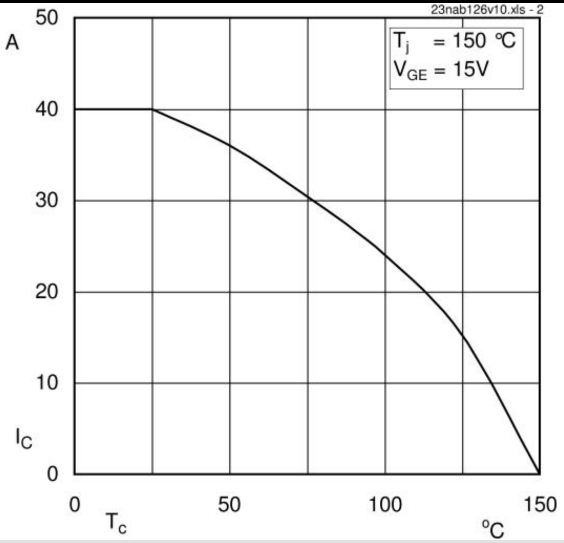


Fig. 2 Typ. rated current vs. temperature

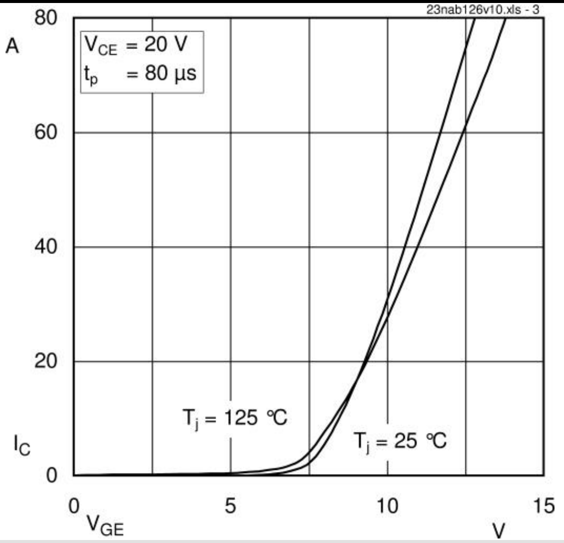


Fig. 3 Typ. transfer characteristic

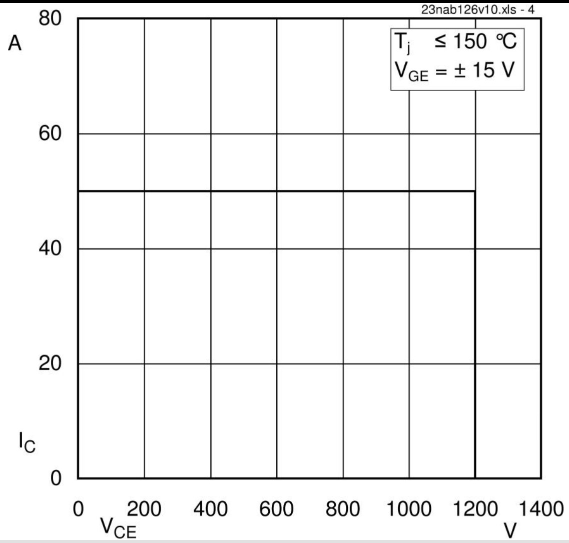


Fig. 4 Reverse bias safe operating area

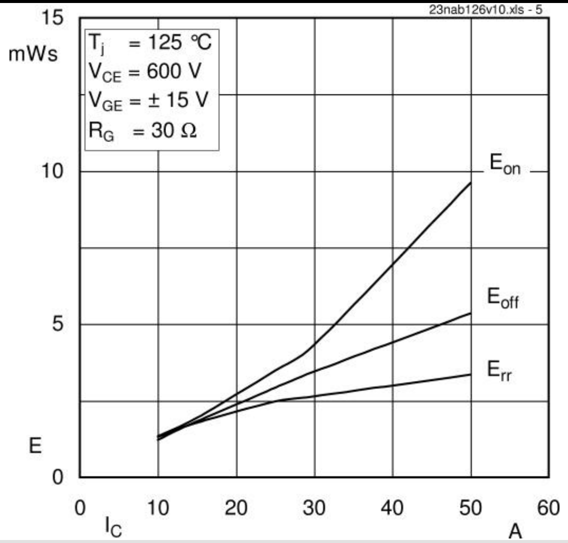


Fig. 5 Typ. Turn-on /-off energy = f (I_C)

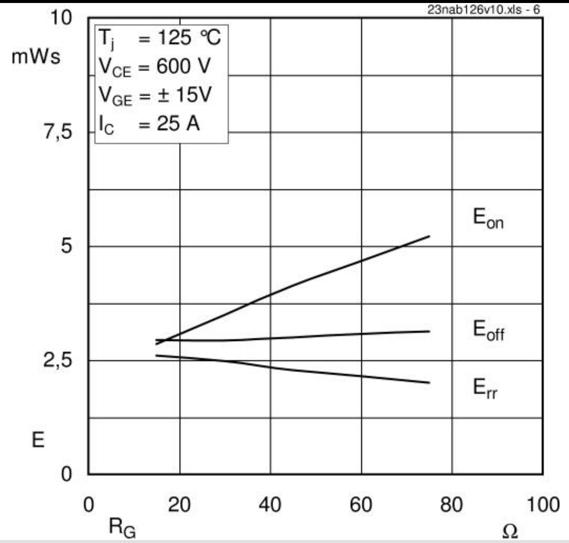


Fig. 6 Typ. Turn-on /-off energy = f (R_G)

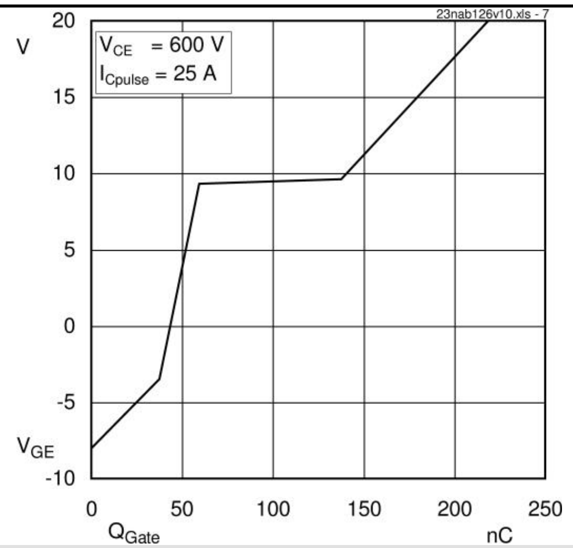


Fig. 7 Typ. gate charge characteristic

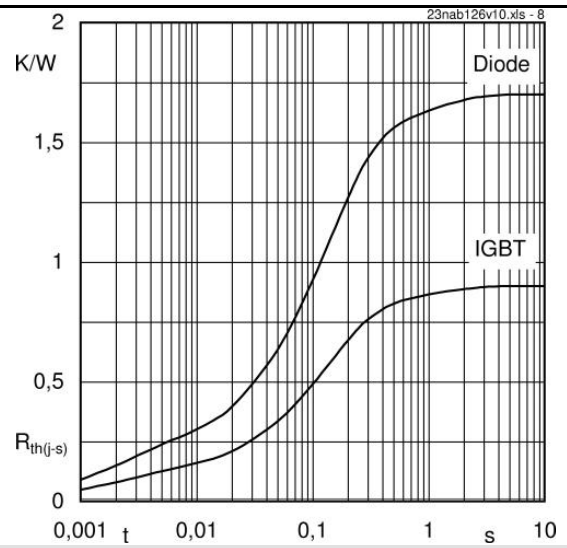


Fig. 8 Typ. thermal impedance

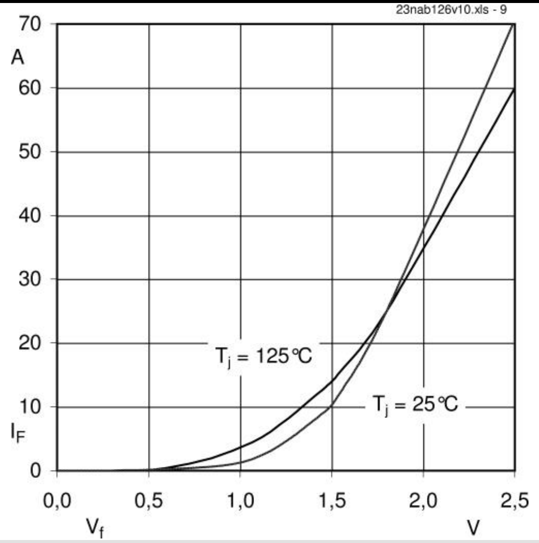


Fig. 9 Typ. freewheeling diode forward characteristic

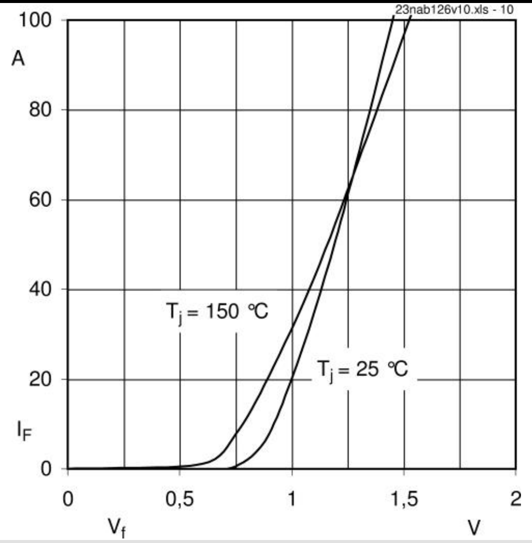
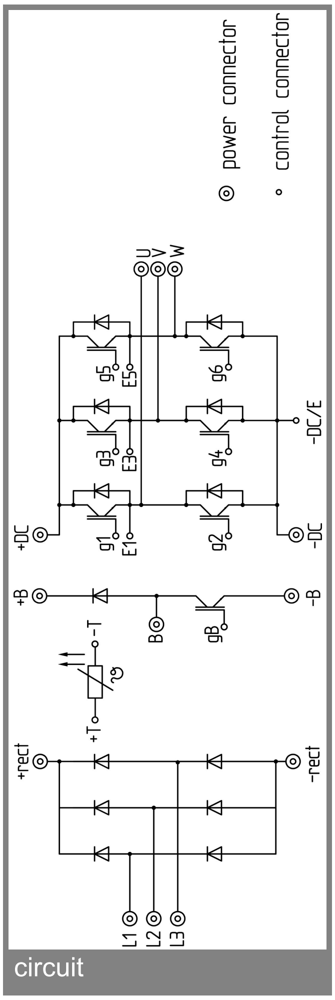
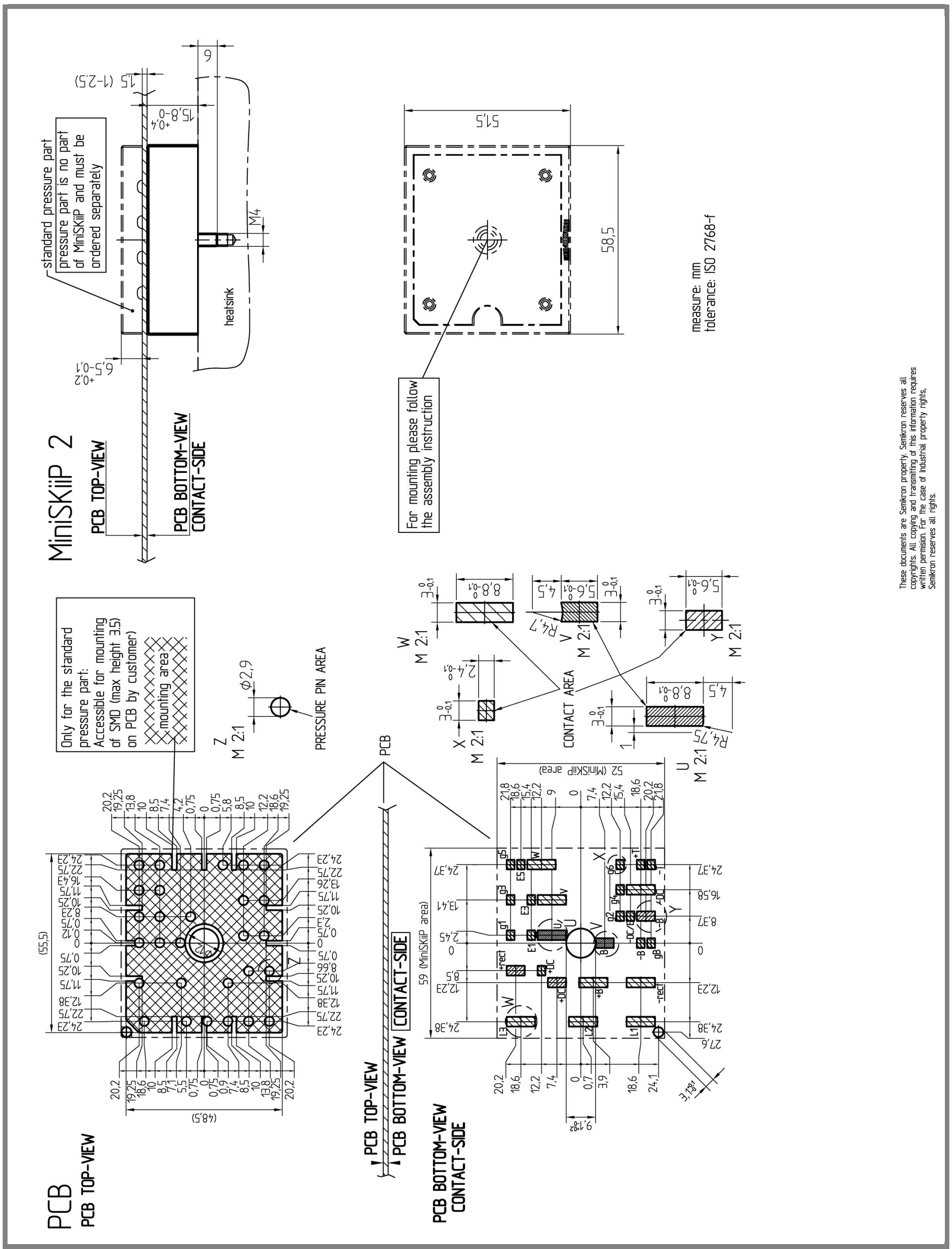


Fig. 10 Typ. input bridge forward characteristic



circuit


MiniSKiiP 2
PCB TOP-VIEW
PCB BOTTOM-VIEW CONTACT-SIDE
standard pressure part
pressure part is no part of MiniSKiiP and must be ordered separately
heatsink
M4
15 (1-2,5)
15,8 +0,4 -0
6,5 +0,2 -0,1
9
51,5
58,5
measure: mm
tolerance: ISO 2768-f
For mounting please follow the assembly instruction
Only for the standard pressure part: Accessible for mounting of SMD (max height 3,5) on PCB by customer)
mounting area
Ø2,9
PRESSURE PIN AREA
Z M 2:1
CONTACT AREA
W M 2:1
V M 2:1
Y M 2:1
X M 2:1
U M 2:1
R4,7
R4,75
PCB
PCB TOP-VIEW
PCB BOTTOM-VIEW CONTACT-SIDE
52 (MiniSKiiP area)
59 (MiniSKiiP area)
(55,5)
(48,5)
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pinout, dimensions

This is an electrostatic discharge sensitive device (ESDS), international standard IEC 60747-1, Chapter IX.

* The specifications of our components may not be considered as an assurance of component characteristics. Components have to be tested for the respective application. Adjustments may be necessary. The use of SEMIKRON products in life support appliances and systems is subject to prior specification and written approval by SEMIKRON. We therefore strongly recommend prior consultation of our personal.